

Customer Segmentation Using Cluster

A PROJECT REPORT

for

AI Project(AI101B)

Session (2024-25)

Submitted by

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**Submitted in partial fulfilment of the
Requirements for the Degree of**

MASTER OF COMPUTER APPLICATION

Under the Supervision of

Mr. Apoorv Jain

Assistant Professor



Submitted to

DEPARTMENT OF COMPUTER APPLICATIONS

KIET Group of Institutions, Ghaziabad

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CERTIFICATE

Certified that **Tannoo Shukla (202410116100220)**, **Tannu Modanwal (202410116100221)**, **Trapti Rajput (202410116100224)**, and **Tanya Singh (202410116100222)** have successfully carried out the project work titled “Customer Segmentation Using Cluster” (Artificial intelligence, AI101B) as part of the curriculum for the Master of Computer Application (MCA) program at Dr. A.P.J. Abdul Kalam Technical University (AKTU) (formerly UPTU), Lucknow, under my supervision.

The project report embodies original work and research undertaken by the students themselves. The contents of the project report do not form the basis for the award of any other degree or diploma to the candidates or any other individual from this or any other university/institution.

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ABSTRACT

Customer segmentation is a crucial strategy in modern marketing that involves dividing a company's customer base into distinct groups with similar characteristics, behaviors, or needs. This project employs unsupervised machine learning techniques, specifically clustering algorithms, to analyze customer data and identify meaningful segments. By using algorithms such as K-Means, customers are grouped based on features like purchase history, spending patterns, demographics, and frequency of visits. The resulting segments enable businesses to tailor marketing strategies, personalize customer experiences, and optimize resource allocation. This approach not only enhances customer satisfaction but also improves overall business efficiency and profitability.

ACKNOWLEDGEMENT

Success in life is never attained single-handedly. My deepest gratitude goes to my project supervisor, **Mr. Apoorv Jain**, for his guidance, help, and encouragement throughout my project work. Their enlightening ideas, comments, and suggestions. Words are not enough to express my gratitude to **Dr. Akash Rajak, Dean and Professor, Department of Computer Applications**, for his insightful comments and administrative help on various occasions. Fortunately, I have many understanding friends, who have helped me a lot on many critical conditions.

Finally, my sincere thanks go to my family members and all those who have directly and indirectly provided me with moral support and other kinds of help. Without their support, completion of this work would not have been possible in time. They keep my life filled with enjoyment and happiness.

Tannoo Shukla
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Trapti Rajput
Tanya Singh

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1. INTRODUCTION

Customer segmentation is a strategic approach that businesses use to divide their customer base into groups with similar characteristics. This segmentation can be based on demographic data, purchase behavior, geographical location, or other attributes.

Clustering, an unsupervised machine learning technique, is especially useful in this context as it does not require labeled data. It identifies inherent patterns in customer data and groups them accordingly.

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2.OBJECTIVE

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6. DATA COLLECTION AND DESCRIPTION

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Customer segmentation is a strategic approach that businesses use to divide their customer base into groups with similar characteristics. This segmentation can be based on demographic data, purchase behavior, geographical location, or other attributes.

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14. BUSINESS IMPLICATIONS

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15. CONCLUSION

Customer segmentation is a strategic approach that businesses use to divide their customer base into groups with similar characteristics. This segmentation can be based on demographic data, purchase behavior, geographical location, or other attributes.

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16. REFERENCES

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CODE AND OUTPUT

1. Importing Required Libraries:

The necessary Python libraries are imported:

- pandas: For data manipulation.
- matplotlib.pyplot and seaborn: For data visualization.
- sklearn.cluster.KMeans: To perform K-means clustering.
- sklearn.preprocessing.StandardScaler: For feature scaling.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
```

2. Loading the Data:

The customer data is loaded from a local CSV file using pandas' read_csv method.

```
# Load data from local CSV file
df = pd.read_csv('/content/customer_data.csv')
print(df.head())
```

3. Displaying the First Few Rows:

df.head() prints the first 5 rows of the dataset to verify the data structure.

```
print(df.head())

# Selecting features for clustering
X = df[['Annual Income (k$)', 'Spending Score (1-100)']]
```

4. Feature Selection:

Only the 'Annual Income (k\$)' and 'Spending Score (1-100)' columns are selected for clustering analysis.

```
# Elbow method to find optimal clusters
inertia = []
k = range(1, 11)
for k in K:
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(X_scaled)
    inertia.append(kmeans.inertia_)
```

5. Feature Scaling:

StandardScaler standardizes features by removing the mean and scaling to unit variance. This is important for K-means as it is distance-based.

```
plt.figure(figsize=(8, 4))
plt.plot(K, inertia, 'bo-')
plt.xlabel('Number of Clusters (k)')
plt.ylabel('Inertia')
plt.title('Elbow Method for Optimal k')
plt.grid(True)
plt.show()
```

6. Elbow Method to Determine Optimal Clusters:

A loop runs K-means clustering for k=1 to k=10 clusters. The inertia (sum of squared distances to the nearest cluster center) is recorded for each k.

```
# Apply KMeans w/ (parameter) n_clusters: int
kmeans = KMeans(n_clusters=5, random_state=42)
df['Cluster'] = kmeans.fit_predict(X_scaled)
```

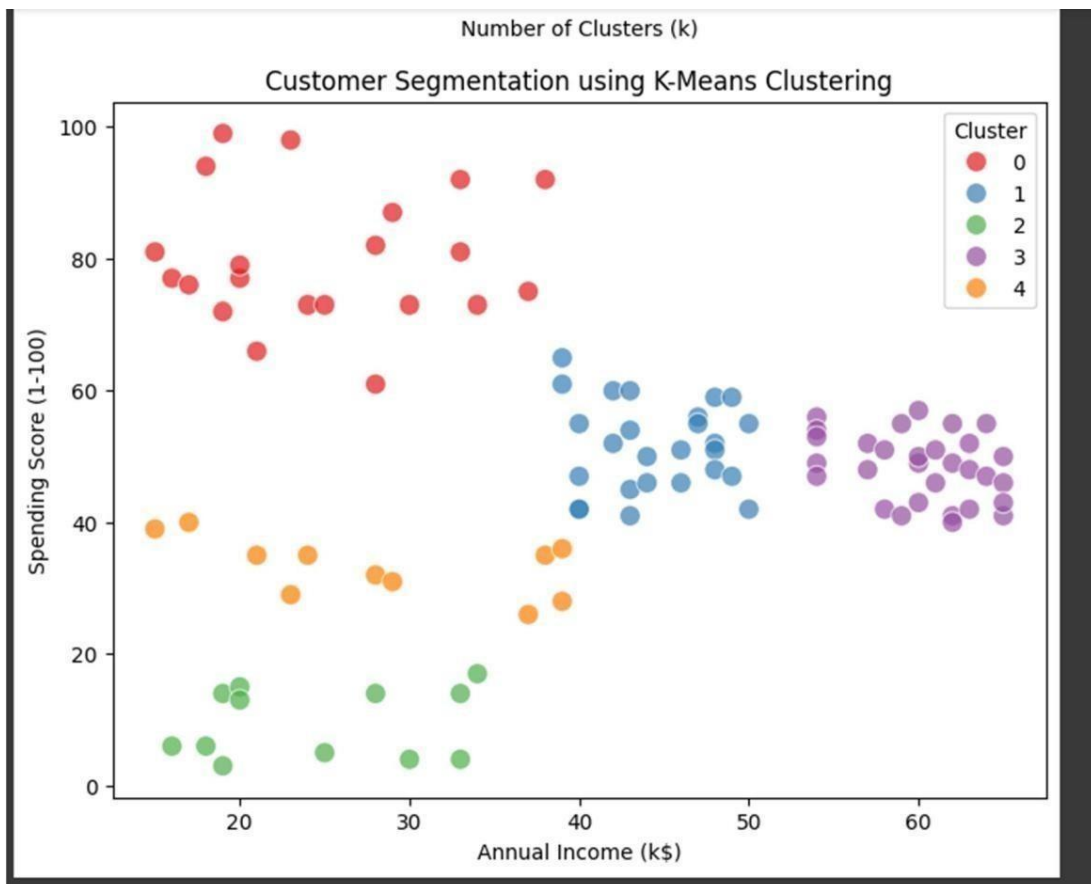
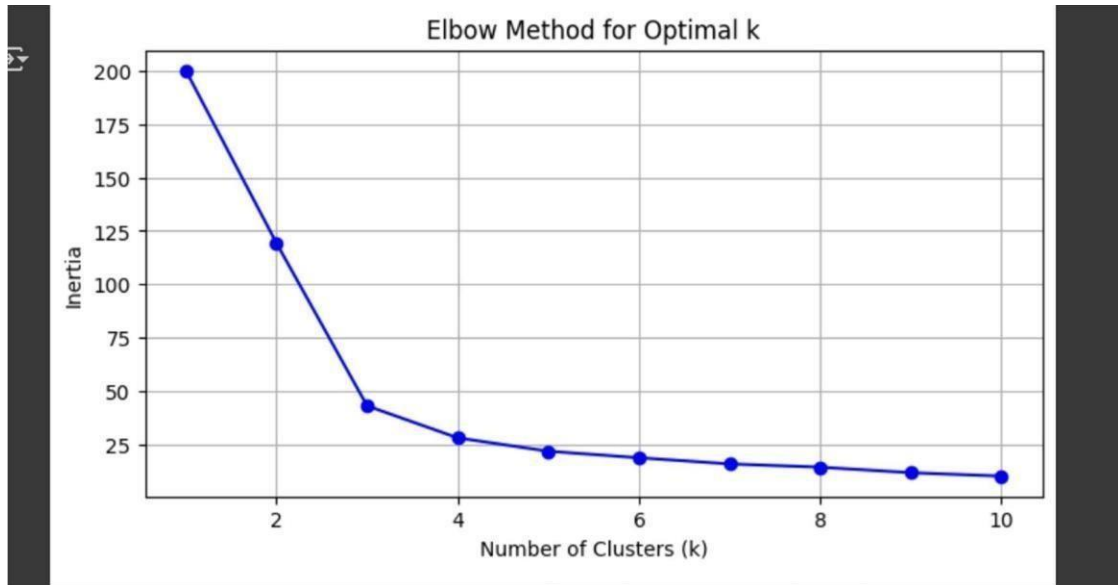
7. Plotting the Elbow Curve:

A line plot is created using matplotlib to visualize the inertia values against different values of k. The 'elbow point' in the plot helps determine the optimal number of clusters.

```
# Visualize clusters
plt.figure(figsize=(8, 6))
sns.scatterplot(x='Annual Income (k$)', y='Spending Score (1-100)',
                hue='Cluster', palette='Set1', data=df, s=100, alpha=0.7)
plt.title('Customer Segmentation using K-Means Clustering')
plt.xlabel('Annual Income (k$)')
plt.ylabel('Spending Score (1-100)')
plt.legend(title='Cluster')
plt.show()
```

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

OUTPUT



Customer Segmentation Using Clustering Techniques

Abstract

Customer segmentation is a crucial element in strategic marketing and customer relationship management. By classifying customers into distinct groups with similar characteristics or behaviors, businesses can tailor their offerings and improve customer satisfaction and profitability. Clustering, an unsupervised machine learning technique, is one of the most effective approaches for customer segmentation. This paper explores various clustering techniques, their implementation in customer segmentation, and a case study demonstrating their effectiveness.

1. Introduction

In the modern competitive business environment, understanding customer behavior and preferences is vital. Customer segmentation enables businesses to divide their customer base into groups that exhibit similar purchasing behaviors, demographics, or psychographics. This segmentation allows for more targeted and effective marketing strategies. Clustering techniques have emerged as powerful tools for uncovering natural groupings within data without prior labeling.

2. Literature Review

Previous studies have shown that customer segmentation leads to improved customer retention and increased revenue. Traditional segmentation methods, such as demographic-based segmentation, are limited in capturing complex customer behaviors. Clustering algorithms, particularly K-Means, Hierarchical Clustering, and DBSCAN, have been widely applied in marketing analytics to overcome these limitations. Recent advancements also incorporate soft clustering approaches like Gaussian Mixture Models (GMM), which assign probabilities to data points belonging to different clusters.

3. Methodology

3.1 Data Collection and Preprocessing

Data used for clustering typically includes demographic information, purchase history, website interaction data, and RFM (Recency, Frequency, Monetary) metrics.

Preprocessing involves handling missing values, scaling features, and encoding categorical variables.

3.2 Feature Selection

Features must be carefully selected to reflect meaningful differences among customers.

Commonly used features include:

- Age
- Annual income - Spending score
- Purchase frequency
- Recency of purchase

3.3 Clustering

Algorithms K-

Means Clustering:

- Partitions data into k clusters by minimizing the variance within each cluster.
- Requires specifying the number of clusters (k) in advance.

Hierarchical Clustering:

- Builds a tree of clusters using either agglomerative or divisive approaches.
- Does not require a pre-specified number of clusters.

DBSCAN (Density-Based Spatial Clustering of Applications with Noise): -

Groups together points that are closely packed and marks points in low-density regions as outliers.

- Useful for discovering clusters of arbitrary shape and identifying noise.

Gaussian Mixture Models (GMM):

- Probabilistic model that assumes data is generated from a mixture of several Gaussian distributions.
- Provides soft clustering by assigning probabilities to cluster membership.

4. Case Study

A dataset of 2,000 retail customers was analyzed using K-Means clustering. After preprocessing and standardizing the data, the Elbow Method was used to determine the optimal number of clusters (k=4). The clusters were profiled as follows:

- Cluster 1: High income, high spending - Target for premium products.
- Cluster 2: Low income, moderate spending - Budget-conscious customers.
- Cluster 3: High frequency, low spending - Loyal but low-value customers.
- Cluster 4: Infrequent, high spenders - Opportunity for loyalty programs.

Visualizations such as scatter plots and heatmaps were used to interpret the clusters.

The segmentation helped the business create targeted marketing campaigns a personalized customer experiences.

5. Evaluation Metrics

- Silhouette Score: Measures how similar an object is to its own cluster versus other clusters.
- Davies-Bouldin Index: Average similarity between clusters, lower values indicate better separation.
- Calinski-Harabasz Index: Ratio of between-cluster dispersion to within-cluster dispersion.

6. Applications and Implications

Clustering-based customer segmentation can lead to:

- Improved marketing efficiency
- Better product recommendations
- Enhanced customer satisfaction
- Reduced churn

Organizations in retail, banking, telecommunications, and e-commerce are increasingly adopting clustering techniques for strategic decision-making.

7. Challenges and Limitations

- Selection of appropriate features and number of clusters
- Sensitivity to outliers and noise
- Interpretability of clusters
- Scalability with large datasets

Future work could explore hybrid models combining clustering with classification algorithms to enhance predictive capabilities.

8. Conclusion

Clustering provides a robust framework for customer segmentation, enabling businesses to uncover hidden patterns in customer behavior. With the right implementation and analysis, clustering can significantly enhance marketing strategies and customer relationship management. As data availability and computational power continue to grow, the role of clustering in business analytics will become even more prominent.

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Customer Segmentation Using Clustering Techniques

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1. Abstract

Effective decisions are mandatory for any company to generate good revenue. In these days competition is huge and all companies are moving forward with their own different strategies. We should use data and take a proper decision. Every person is different from one another and we don't know what he/she buys or what their likes are. But, with the help of machine learning technique one can sort out the data and can find the target group by applying several algorithms to the dataset. Without this, It will be very difficult and no better techniques are available to find the group of people with similar character and interests in a large dataset. Here, The customer segmentation using K-Means clustering helps to group the data with same attributes which exactly helps to business the best. We are going to use elbow method to find the number of clusters and at last we visualize the data.

2. Keywords

Clustering, Elbow Method, K-Means Algorithm, Customer Segmentation, Visualization.

3. Introduction

3.1 Introduction

Nowadays the competition is vast and lot of technologies came into account for effective growth and revenue generation. For every business the most important component is data.

With the help of grouped or ungrouped data, we can perform some operations to find customer interests.

Data mining helpful to extract data from the database in a human readable format. But, we may not know the actual beneficiaries in the whole dataset. Customer Segmentation is useful to divide the large data from dataset into several groups based on their age, demographics, spent, income, gender, etc. These groups are also known as clusters. By

this, we can get to know that, which product got huge number of sales and which age group are purchasing etc. And, we can supply that product much for better revenue generation.

Initially we are going to take the old data. As we know that old is gold so, by using the old data we are going to apply K-means clustering algorithm and we have to find the number of clusters first. So, at lastly, we have to visualize the data. One can easily find the potential group of data while observing that visualization.

The goal of this paper is to identify customer segments using the data mining approach, using the partitioning algorithm called as K-means clustering algorithm. The elbow method determines the optimal clusters.

3.2 Problem Statement

Customer Segmentation is the best application of unsupervised learning. Using clustering, identify segments of customers in the dataset to target the potential user base. They divide customers into various groups according to common characteristics like gender, age, interest, and spending habits so they can market to each group effectively. Use K-Means Clustering and also visualize the gender and age distributions. Then analyze their annual income and spending scores. As it describes about how we can divide the customers based on their similar characteristics according to their needs by using k-means clustering which is a classification of unsupervised machine learning.

4. Existing System

The existing method is storing customer data through paperwork and computer software (digital data) is increasing day by day. At end of the day they will analyse their data as how many things are sold or actual customer count etc. By analysing the collected data they got to know who is beneficial to their business and increase their sales. It requires more time and

more paperwork. Also, it is not much effective solution to find the desired customers data.

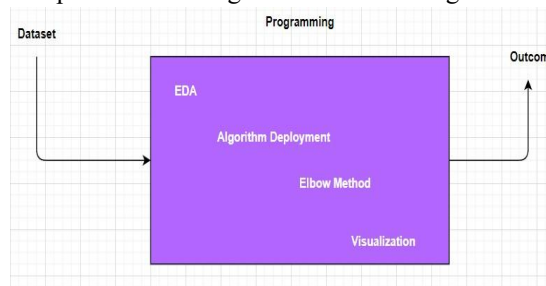
5. Proposed System

5.1 Proposed Method

To overcome the traditional method i.e paper work and computerized digital data this new method will play vital role. As we collect a vast data day by day which requires more paperwork and time to do. As new technologies were emerging in today's world. Machine Learning which is powerful innovation which is used to predict the final outcome which has many algorithms. So for our problem statement we will use K-Means Clustering which groups the data into different clusters based on their similar characteristics. And then we will visualize the data.

5.2 System Architecture

Initially we will see the dataset and then we will perform exploratory data analysis which deals with the missing data, duplicates values and null values. And then we will deploy our algorithm k-means clustering which is unsupervised learning in machine learning.



As in order to find the no of clusters we use elbow method where distance will be calculate through randomly chosen centres and repeat it until there is no change in cluster centres. Thereafter we will analyse the data through data visualization. Finally we will get the outcome.

5.3 Algorithm

- K Means algorithm in an iterative algorithm that tries to partition the dataset into K predefined distinct non overlapping sub groups which are called as cluster.
- Here K is the total no of clusters.
- Every point belongs to only one cluster.
- Clusters cannot overlap.

5.3.2 Steps of Algorithm

- Arbitrarily choose k objects from D as the initial cluster centers.
- Repeat.
- Assign each object to the cluster to which the object is the most similar, based on the mean value of the objects in the cluster.
- Update the cluster means, i.e. calculate the mean value of the objects for each cluster.
- Until no change.

6. Methodology

1. First of all we will import all the necessary libraries or modules (pandas, numpy, seaborn).
2. Then we will read dataset and anyalse whether it contains any null values, missing values and duplicate values. So we will fix them by dropping or fixing the value with their means, medians etc which is technically named as Data Preprocessing.
3. We will deploy our model algorithm K-Means Clustering, which divides the data into group of clusters based on similar characteristics. To find no.of clusters we will use elbow method.
4. Finally, we will visualize our data using matplotlib, which concludes the customers divided into groups who are similar to each other on their group.

5.3.1 K-Means Clustering

7. Implementation And Analysis

7.1 Overview of a Dataset

This is a mall customer segmentation data which contains 5 columns and 200 rows.

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40
...	...	...	...	...	...
195	196	Female	35	120	79
196	197	Female	45	126	28
197	198	Male	32	126	74
198	199	Male	32	137	18
199	200	Male	30	137	83

200 rows × 5 columns

7.2 Exploratory Data Analysis

It deals with the data preprocessing, whether it contains any missing values or null values. There after we will see the information and description of the dataset.

7.2.1 Information of the dataset

#df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  -
0   CustomerID            200 non-null   int64
1   Gender                200 non-null   object
2   Age                   200 non-null   int64
3   Annual Income (k$)    200 non-null   int64
4   Spending Score (1-100) 200 non-null   int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

As here it overview the information of the data. And it gives it doesn't contain any null values.

As we will remove the irrelevant data which is customer id.

```
df.drop(["CustomerID"],axis=1, inplace=True)
```

```
# so here customer data is not required to our analysis. We will drop it.
df.drop(["CustomerID"], axis=1, inplace=True)

# printing data frame again (Now, CustomerID column is removed)
df
```

	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	Male	19	15	39
1	Male	21	15	81
2	Female	20	16	6
3	Female	23	16	77
4	Female	31	17	40
5	Female	22	17	76
6	Female	35	18	6
7	Female	23	18	94

7.2.2 Description of the data

#df.describe()

	Age	Annual Income (k\$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000
mean	38.850000	60.560000	50.200000
std	13.969007	26.264721	25.823522
min	18.000000	15.000000	1.000000
25%	28.750000	41.500000	34.750000
50%	36.000000	61.500000	50.000000
75%	49.000000	78.000000	73.000000
max	70.000000	137.000000	99.000000

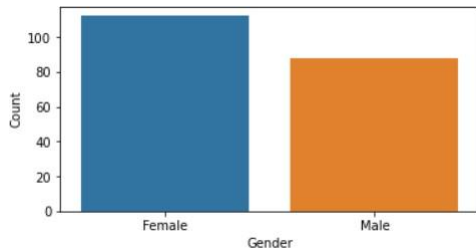
It describes about the count which counts the no of rows in it, mean of the columns, standard deviations, maximum and minimum and percentiles etc.

7.3 Gender plot Analysis

Here it overview the gender analysis

```
#Gender Distribution
genders=df.Gender.value_counts()
plt.figure(figsize=(6,3))
sns.barplot(x=genders.index,y=genders.values)
plt.xlabel('Gender')
plt.ylabel('Count')
plt.show
```

So we label the x-axis as Gender and y-axis as Count and we plot it by using barplot.



From the plot we will conclude that there

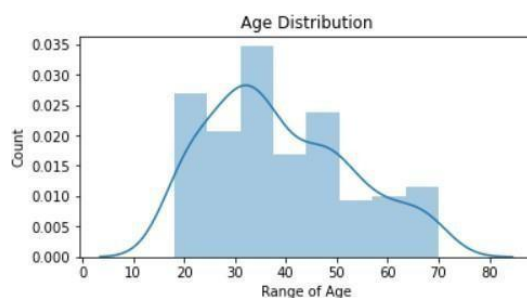
are more female customers than the male customers i.e. female customers are more than 100 whereas male customers are nearly 80.

7.4 Age plot

We will use distplot for the distribution of age of the customers.

```
plt.figure(figsize=(6,3))
sns.distplot(df['Age'])
plt.title('Age Distribution')
plt.xlabel('Range of Age')
plt.ylabel('Count')
plt.show()
```

So we label X-axis as range of age and y-axis as count.

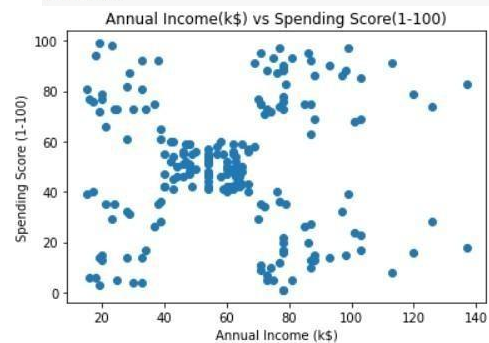


From the plot, it varies the age from nearly 20 to 70. It is evident that the age of the customers between 30 - 40 are more, then after 20-30 etc.

7.5 Annual Income vs Spending Score

As we will use scatterplot and labelled x-axis as Annual Income(k\$) and y-axis as Spending Score(1-100)

```
plt.scatter(df['Annual Income (k$)'],df['Spending Score (1-100)'])
plt.title('Annual Income(k$) vs Spending Score(1-100)')
plt.xlabel('Annual Income (k$)')
plt.ylabel('Spending Score (1-100)')
plt.show()
```



From the plot we observed that it varies from low annual income with low expenditure or spending money to high annual income with high expenditure.

7.6 Elbow Method

The elbow method is based on the observation that increasing the number of clusters can help to reduce the sum of within-cluster variance of each cluster. This is because having more clusters allows one to capture finer groups of data objects that are more similar to each other. To define the optimal clusters, firstly, we use the clustering algorithm for various values of k.

This is done by ranging k from 1 to 10 clusters. Then we calculate the total intra-cluster sum of square. Then, we proceed to plot intra-cluster sum of square based on the number of clusters. The plot denotes the approximate number of clusters required in our model. The optimum clusters can be found from the graph where there is a bend in the graph.

First we will consider the data X which has only two columns: they are annual income and spending score.

```
X=df[['Annual Income (k$)','Spending Score (1-100)']]
```


X.head()

	Annual Income (k\$)	Spending Score (1-100)
0	15	39
1	15	81
2	16	6
3	16	77
4	17	40

wcss=[]

for i in range(1,11):

 km=KMeans(n_clusters=i)

 km.fit(X)

 wcss.append(km.inertia_)

plt.figure(figsize=(6,3))

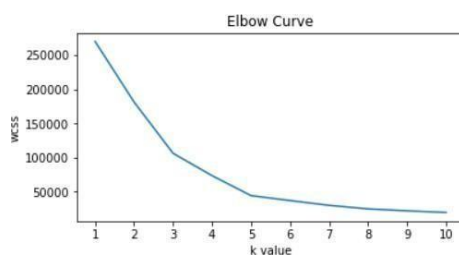
plt.plot(range(1,11),wcss)

plt.title('Elbow Curve') plt.xlabel('k
value')

plt.xticks(np.arange(1,11,1))

plt.ylabel('wcss')

plt.show()



So from the graph we observed that the at 5 there is bend and it can be considered as k which is no of clusters.

Therefore, k=5 i.e no of clusters are equal to 5.

7.7 Fitting the Algorithm

```
km=KMeans(n_clusters=5)
km.fit(X)
y=km.predict(X)
df['cluster']=y
df.head()
```

As here we initialized the kmeans as km with 5 clusters and we will fit it. There after we will predict the data and store it in y. And then we will add new column named as Cluster and data as y.

	Gender	Age	Annual Income (k\$)	Spending Score (1-100)	Cluster
0	Male	19	15	39	4
1	Male	21	15	81	3
2	Female	20	16	6	4
3	Female	23	16	77	3
4	Female	31	17	40	4

So from the figure we observed that each customer is labelled with cluster which is based on their characteristics.

7.8 Visualization the clusters

Visualizing the clusters based on Annual Income and Spending Score of the customers. As here we plot a graph named as Clusters of Customers to visualize the data in terms of groups or cluster. plt.figure(figsize=(15,7))

```
plt.scatter(df["Annual Income (k$)"][df.Cluster == 0], df["Spending Score (1-100)"][df.Cluster == 0], c='blue', s=60, label='Cluster 0')
```

```
plt.scatter(df["Annual Income (k$)"][df.Cluster == 1], df["Spending Score (1-100)"][df.Cluster == 1], c='red', s=60, label="Cluster 1")
```

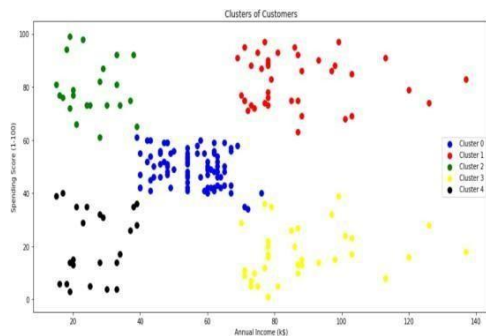
```
plt.scatter(df["Annual Income (k$)"][df.Cluster == 2], df["Spending Score (1-100)"][df.Cluster == 2], c='green', s=60, label='Cluster 2')
```

```
plt.scatter(df["Annual Income (k$)"][df.Cluster == 3], df["Spending Score (1-100)"][df.Cluster == 3], c='yellow', s=60, label='Cluster 3')

plt.scatter(df["Annual Income (k$)"][df.Cluster == 4], df["Spending Score (1-100)"][df.Cluster == 4], c='black', s=60, label='Cluster 4')

plt.title('Clusters of Customers')

plt.legend() plt.xlabel('Annual Income (k$)') plt.ylabel('Spending Score (1-100)') plt.show()
```



So from the above one we observed that there are 5 clusters which are named as 0, 1, 2, 3, 4.

- Cluster 0 which is at centre, average annual income with average spending score.
- Cluster 1 which is at top right, highest annual income with highest spending score.
- Cluster 2 which is at top left, lowest annual income with highest spending score.
- Cluster 3 which is at bottom right, high annual income with low spending score.

8. Conclusion

So we concluded that the ,

- The Highest income , high spending can be target these type of customers as they earn more money and spend as much as they want.
- Highest income, low spending can be target these type of customers by asking feedback and advertising the product in a better way.
- Average income, Average spending may or may not be beneficial to the mall owners of this type of customers.
- Low income, High spending can be target these type of customers by providing them with low-cost EMI's etc.
- Low income, Low spending don't target these type of customers because they earn a bit and spend some amount of money.

So high income, high spending are the most beneficial ones to the mall owners which increases the owner's business. (Cluster 1)

- Cluster 4 which is at bottom left, lowest annual income with lowest spending score.

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